

DETECTION OF PARKINSONIAN SYNDROME USING WEARABLE DEVICE



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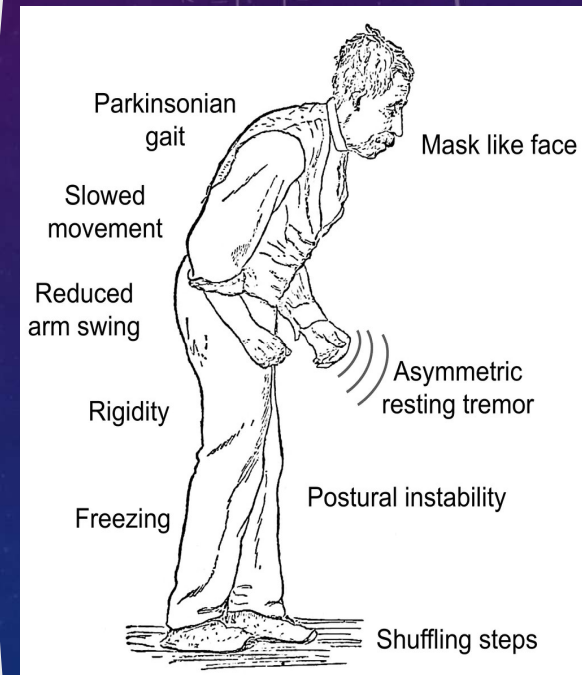
OBJECTIVE

To develop a unique system that is wearable, cost-effective, simple to operate and can test main symptoms of Parkinson's disease which are Hand Tremors and Parkinsonian Gait.

To help enhance the diagnosis, treatment, and research of patients with this disease.

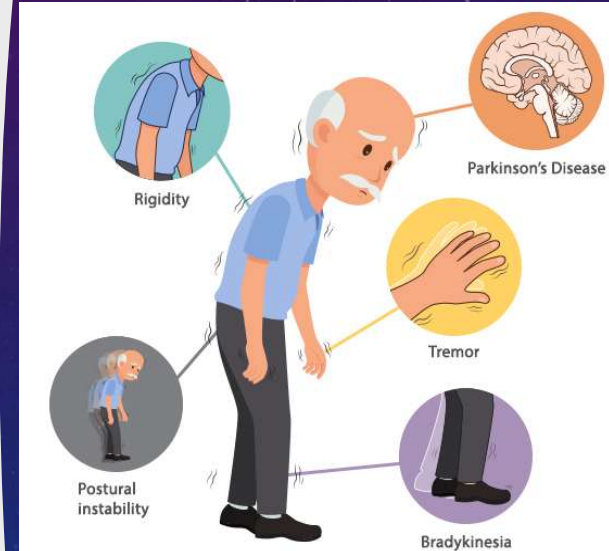
INTRODUCTION

- ❑ Parkinson's disease, mostly found in older people, is one of the chronic neurodegenerative diseases which exhibits a progressive loss of the ability to coordinate movements
- ❑ Symptoms of Parkinson's disease that can be detrimental to the quality of life of patients and are often disabling include, bradykinesia, resting tremor, depression and cognitive impairment, rigidity, and postural instability .
- ❑ There is an immediate requirement in the medical field for early detection and monitoring of Parkinson's disease and there is a need for the development of diagnostic tools to characterize progressive movement disorders.
- ❑ It is important to test hand tremors and at the same time perform gait analysis. This has led to the implementation of systems that can monitor the symptoms in a non-controlled environment



INTRODUCTION

- ❑ Patients may not be comfortable in using huge machines with sophisticated features especially when it is known that the largest risk factor is the age of an individual for the development and progression of this disease .
- ❑ Hence, a simple and easy-to-use system is proposed in this research paper which can test the presence of Gait and Tremors within minutes. Further, these outputs can be used by physicians to provide a further diagnosis.
- ❑ These outputs can also be uploaded on the cloud and auto-emailed to consulting physicians if the patient desires.
- ❑ This system is not only cost-effective and comprehensive but also is designed in a way that people of all age groups can handle the wearable device comfortably.



LITERATURE REVIEW

Author Names with year	Input	Output	Results
Christopher A Ross , Wanli W Smith 2007	Current Research models for Parkinson's Disease	Review on which model would replicate all the features.	None of the single models replicate all the features of PD. Genetic models in combination with toxins or other environmental manipulation may provide better models of PD pathogenesis.
A Schrag, M Jahanshahi, N Quinn 2000	Answers of Patients for disease-specific QoL questionnaire (PDQ-39) and the Beck depression inventory.	Factors that determine quality of life (QoL) in patients with idiopathic Parkinson's disease in a population-based sample	Depression, disability, postural instability, and cognitive impairment have the greatest influence on QoL in Parkinson's disease. The improvement of these features should therefore become an important target in the treatment of the disease.
Annese, V.F., Mezzina, G., Gallo, V.L., Scarola, V. and De Venuto, D. 2017	1-bit signals are used to define muscular indexes and get muscular information.	Proposed platform for the quantification of PD symptoms	The system has been tested on 4 subjects: 2 affected by PD and 2 healthy subjects (control group). The experimental results highlight the validity of the proposed solution in Disease recognition and the outcomes match the clinical literature results.
A Bermeo, M Bravo, M Huerta, A Soto 2016	Data tremors acquired by the sensor device	An Android application which allows an evaluation of the state of PD patients based on three types of tests that are in the Unified Parkinson's Disease Rating Scale	Design and development of a system to monitor tremors in patients with Parkinson's disease based on Arduino open-source prototyping platform

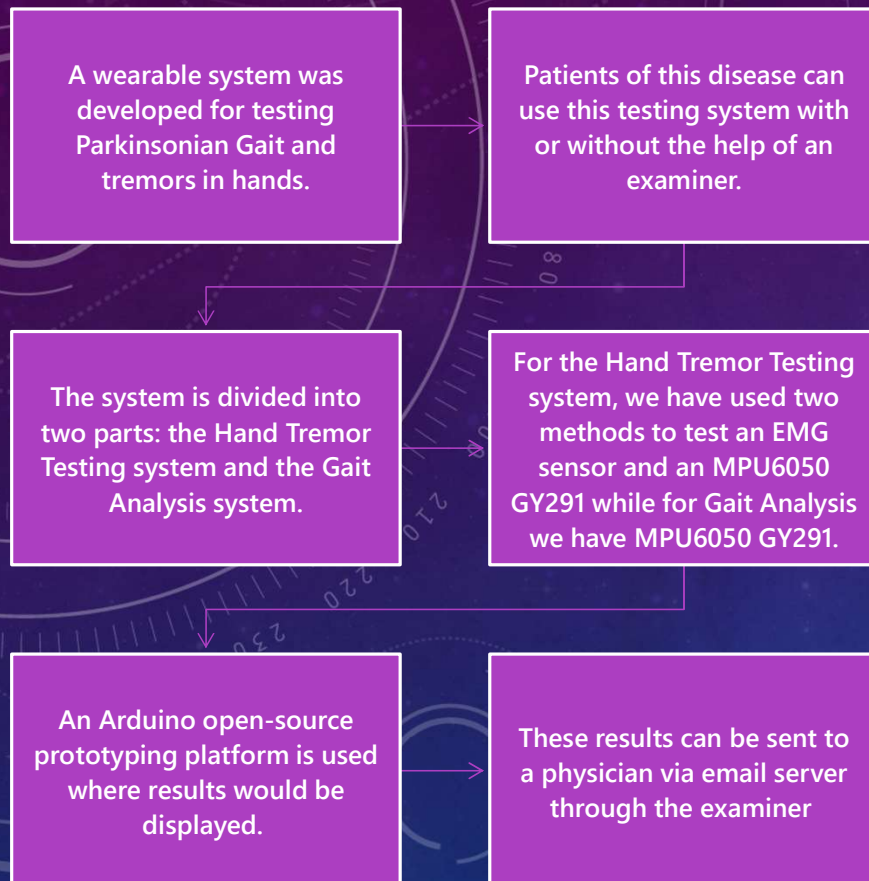
LITERATURE REVIEW

Author Names with year	Input	Output	Results
Thomas Lennon, Trevor Bernier, Daniel Tamayo ,Cody Goldberg , Kunal Mankodiya 2015	Data of dyskinesia	Intelligent wristband that has motion sensors to collect and send high quality data of dyskinesia	Multi-sensory system for monitoring dyskinesia disorders was developed so that physicians can get insight into the everyday symptoms of their patients
Blake P Printy, Lindsey M Renken, John P Herrmann, Isac Lee, Bryant Johnson, Emily Knight, Georgeta Varga, Diane Whitmer 2014	Kinematic data from a small cohort of PD patients during guided movement task	A smartphone application that could quantify the severity of Parkinson's disease (PD) motor symptoms	Using a support vector machine classifier, a classification accuracy of 0.945 was achieved under 6-fold cross validation. Accurate classification for discriminating between more severe and less severe bradykinesia was not achieved with these methods
D De Venuto I, V.F.Annese, G. Defazio, V.L. Gallo I, G. Mezzina 2017	Responses before and after the drug (Levodopa) treatment	Cyber-physical system for real-time processing EEG and EMG signals.	The system quantifies the effect of the Levodopa treatment detecting. Brain implications shows a mean reduction of 5% on the evaluated potentials.
John . V. Hindle 2010	Data of Patients suffering from Parkinson disease ,their age and other diseases they might be suffering from	Summary on how age affects Parkinson disease patients	Ageing may affect the clinical presentation of PD with altered drug side effects, increased risk of developing dementia and an increased likelihood of admission to a nursing home.

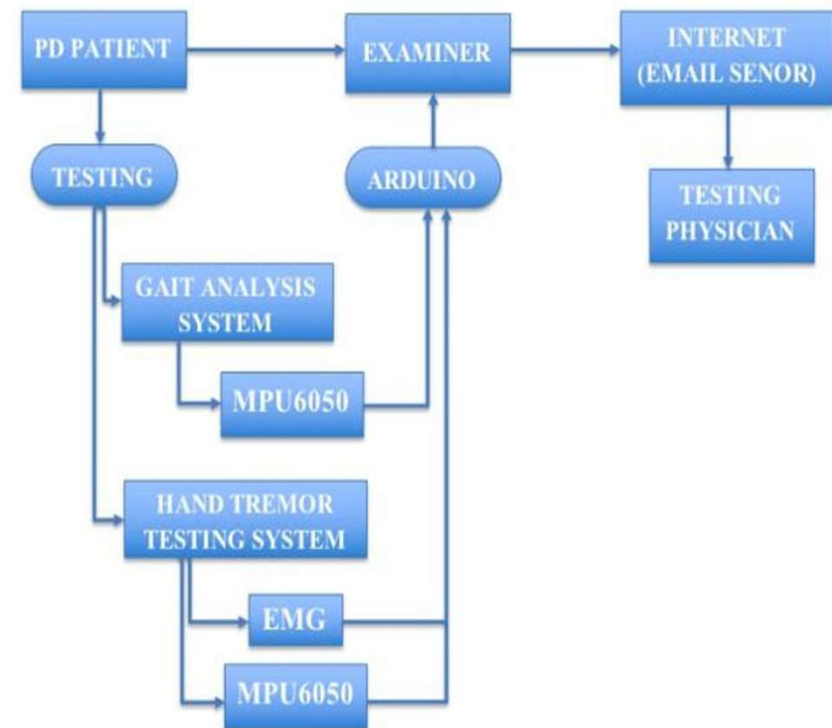
LITERATURE REVIEW

Author names with year	Input	Output	Results
Arif Reza Anwary, Hongnian Yu, Michael Vassallo 2018	Accelerometer data, gyroscope data	An android app to collect real-time synchronous sensor output	This paper has proposed an automatic gait features extraction method to analyze the data for stride number, distance, speed, length and period of stride, stance, and swing phases during walking.
di Biase L, Di Santo A, Caminiti ML, De Liso A, Shah SA, Ricci L, Di Lazzaro V 2020	Data of Gait analysis in Parkinson's disease	Review on the most relevant technologies used to evaluate gait features and the associated algorithms	Only a limited number of studies reported algorithms that were accurate enough deemed to be useful for diagnosis and symptoms monitoring. In addition, none of the reported algorithms and technologies has been validated in large scale, independent studies.
Whittle M.W 2007		Overview of some of the ways in which gait analysis is used.	The requirements for accuracy are generally not as great in the clinical setting as they are in the research laboratory, so long as the measurement errors are not large enough to cause a misinterpretation of the clinical condition

METHODOLOGY



Block Diagram Of Proposed System

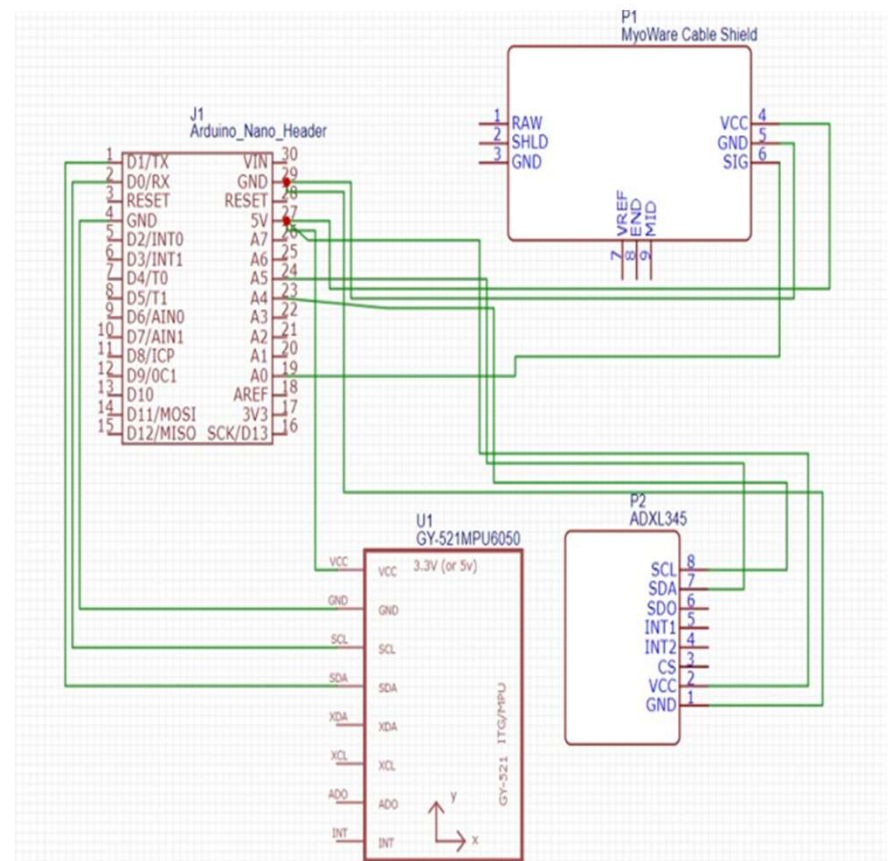


SCHEMATIC

A system made up of EMG Sensor, Arduino Nano, Biomedical sensor pads, connecting wires and the USB cable is used for the detection of tremors. The system would be connected to the patient's hand

The second testing system is made up of MPU6050 GY291, Arduino Nano, connecting wires, and USB cable. This system would be connected to the finger of the patient through a glove.

The Gait Analysis system consists of an Arduino Nano and an MPU6050 sensor (three-axis accelerometer and three-axis gyroscope). This is used for acceleration sensing, calculating the pitch angle of the leg's swing, and transmitting real-time signals.



RESULTS

Hand Tremors-EMG Sensor

- ❑ EMG sensor is used along with Arduino open-source prototyping software to measure hand tremors using EMG Sensor.
- ❑ In Serial Monitor and Plotter we would be able to observe the values and graphs obtained, respectively.



Fig. 1. Hardware of the designed system for testing tremors

- ❑ The obtained values of sensor values and mV values on Serial Monitor

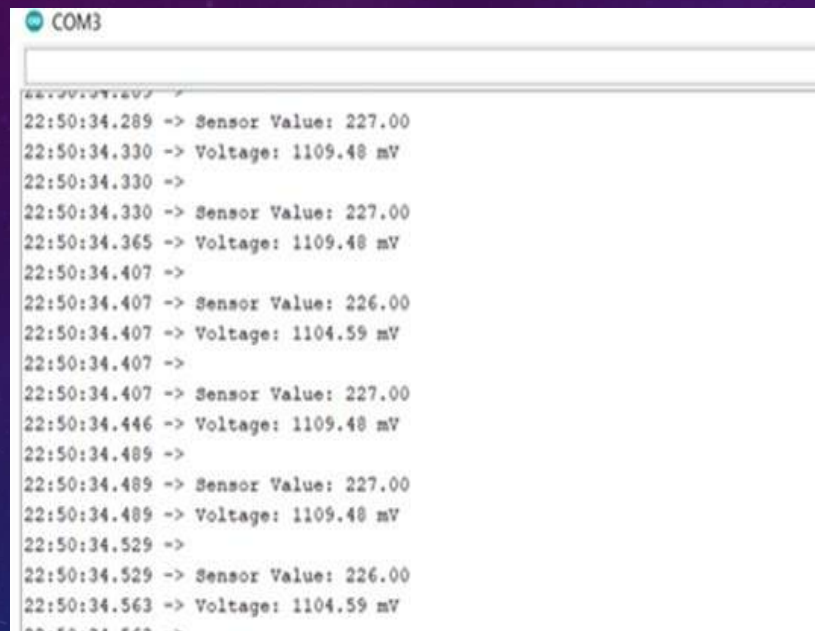


Fig. 2. Sensor value and mV value as displayed on Serial Monitor

- ❑ Graph plotted helps in identification of Parkinson's disease .



Fig 3. Graph of value obtained and mV used when a test run was conducted on the entire system

Hand Tremors-MPU6050 GY291

- ❑ MPU6050 GY291 is used along with Arduino open source prototyping software to measure hand tremors using EMG Sensor.
- ❑ Patient has to place the sensor on their finger so as to obtain the values of acceleration along all three axes

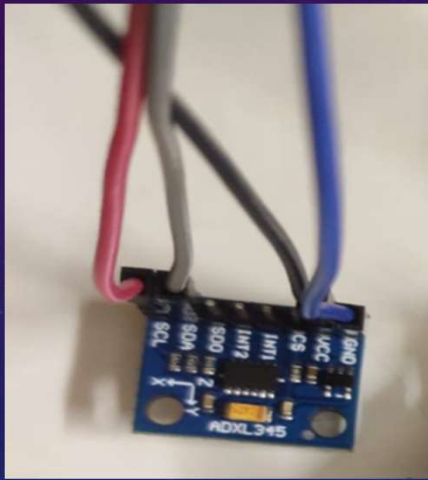


Fig. 4. MPU6050 GY291

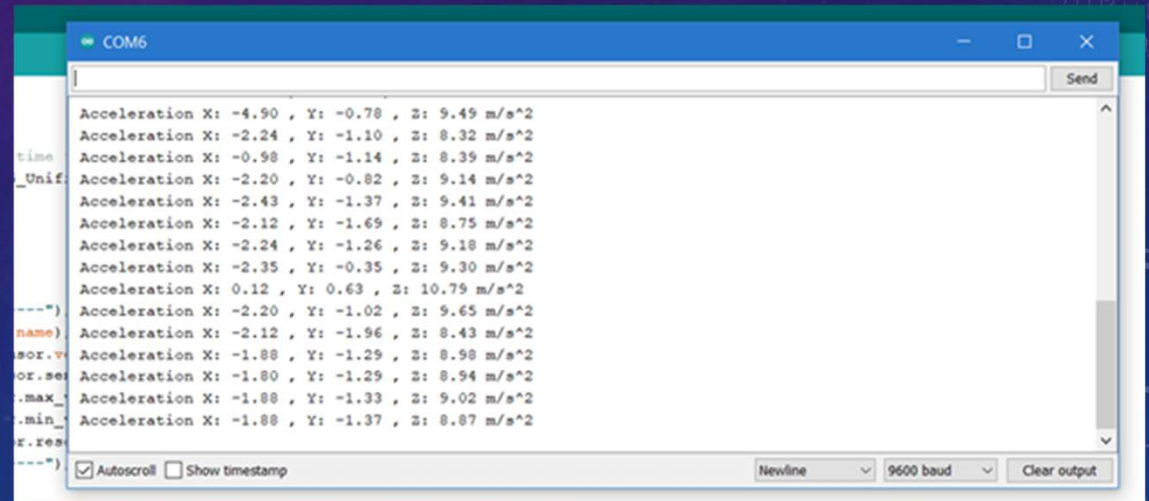


Fig. 5. Acceleration values in all three axes as displayed on Serial Monitor

- Testing Values obtained after running the system for detecting hand tremors on a person with no history of Parkinson's disease

Testing Values

HAND TREMOR		SENSOR NAME			
VALUE NUMBER	EMG	MPU 6050			
	Sensor Value	mV	x(m/s^2)	y(m/s^2)	z(m/s^2)
1	227	1109.48	-1.06	-0.27	9.22
2	227	1109.48	-1.1	0.31	8.51
3	226	1104.59	-2.12	0.16	9.89
4	227	1109.4	-2.24	-0.63	19.22
5	227	1109.48	0.51	0.43	4.67
6	226	1104.59	-0.98	0.24	7.88
7	226	1104.59	-1.45	1.06	9.1
8	226	1104.59	-2.28	-0.98	9.49
9	225	1099.71	-3.22	-1.41	13.38
10	227	1108.48	-2.12	2.2	7.96
AVERAGE	226.4	1106.439	-1.606	0.111	9.932

Fig. 7. Values for Hand Tremors when tested on healthy Patient A

Fig. 6. Test Values for Hand Tremors System

Gait Analysis -MPU6050

- ❑ MPU6050 is used along with Arduino open-source prototyping software to perform Gait Analysis.
- ❑ Gait parameters can be obtained: Euler angle, Pitch, Yaw, Roll, Quaternion values, Acceleration, and World-frame Acceleration values. These can be further processed into a quantitative study which would help detect Parkinson's Disease at an early stage on Serial Monitor and Serial Plotter

```

Trail MPU areal -13 269 -1279
#include "I2C" 123.38 8.21 179.45
#include "MPU6050" -14 273 -1281
#if I2CDEV_I2C 123.43 8.22 179.46
#include "I2CDEV_I2C" -13 273 -1278
#endif 123.47 8.21 179.47
MPU6050 mpu; areal -20 271 -1271
//define OI 123.52 8.21 179.47
#define OUT -19 269 -1272
#define OUT 123.57 8.22 179.48
//define OI -20 269 -1267
//define OI 123.62 8.21 179.49
areal -19 270 -1259
#define LED 123.67 8.21 179.49
bool blinkSt -19 277 -1268
euler 123.72 8.21 179.51
// MPU contr
bool dmRea
  
```

Fig. 7. Euler angles and acceleration values displayed on serial monitor

```

Trail MPU ypr 24.63 4.83 -6.22
#include "I2C" -0.06 0.97 0.21 0.03
#include "MPU6050" 24.68 4.83 -6.22
#if I2CDEV_I2C -0.06 0.97 0.21 0.03
#include "I2CDEV_I2C" 24.73 4.83 -6.22
#endif -0.06 0.97 0.21 0.03
MPU6050 mpu; ypr 24.77 4.83 -6.22
#define OUT -0.06 0.97 0.21 0.03
//define OI 24.82 4.83 -6.23
//define OI -0.06 0.97 0.21 0.03
//define OI 24.87 4.83 -6.23
#define OUT -0.06 0.97 0.21 0.03
ypr 24.92 4.83 -6.23
#define LED -0.06 0.97 0.21 0.03
bool blinkSt 24.97 4.83 -6.22
quat -0.06 0.97 0.22 0.03
// MPU contr
bool dmRea
  
```

Fig. 8. Yaw, Pitch, Roll and Quaternion values displayed on serial monitor

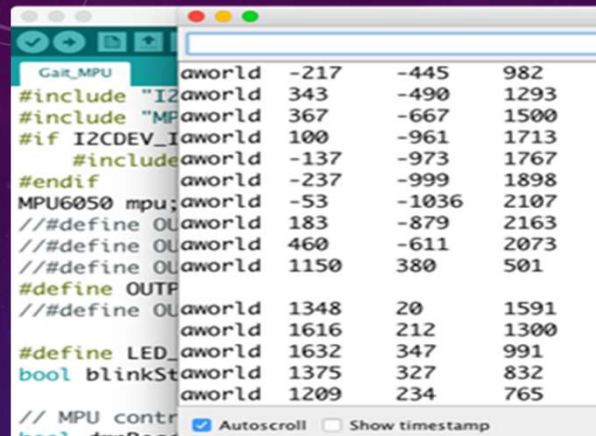


Fig. 9. World acceleration values displayed on serial monitor

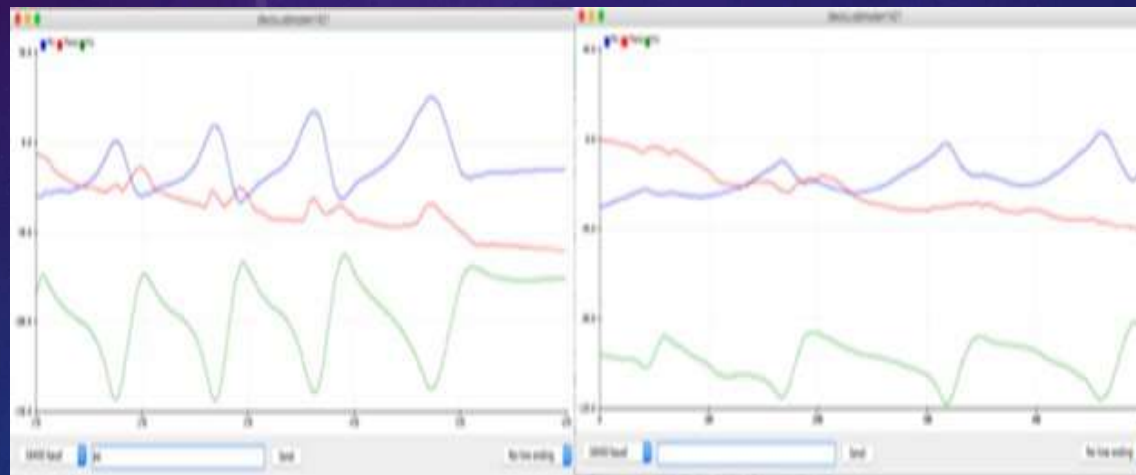


Fig. 10. Results of a Healthy Person vs a Patient – Euler Angles (Phi, Theta, Psi)

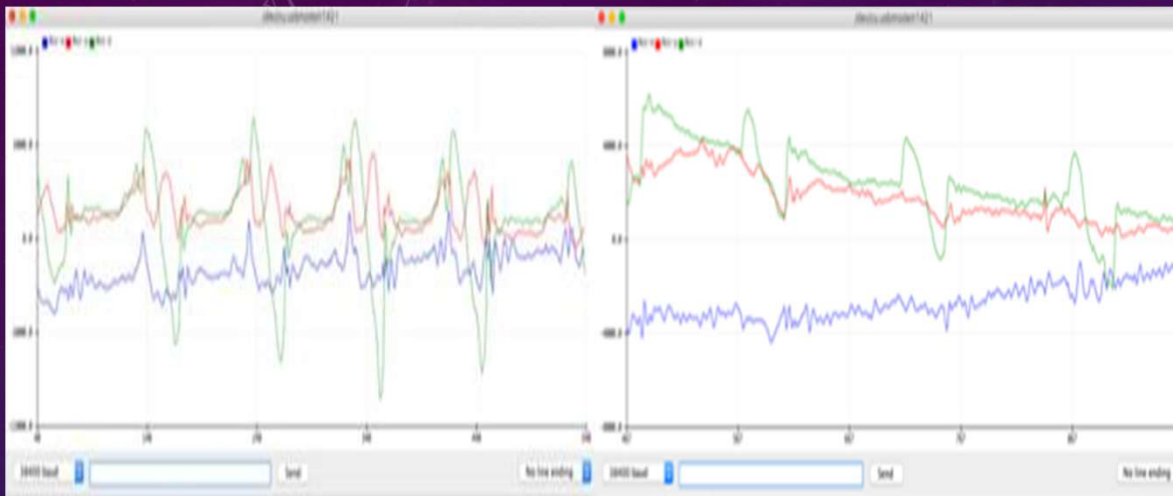


Fig. 11. Results of a Healthy Person vs a Patient – Acc with respect to x, y, z - axes

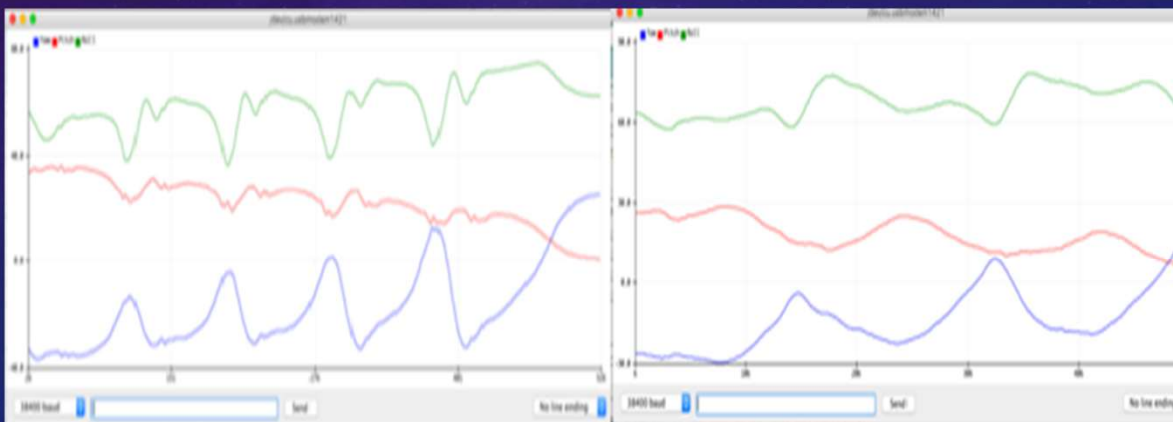


Fig. 12. Results of a Healthy Person vs a Patient – Yaw, Pitch, Roll

CONCLUSION

- ❑ This Paper describes a method for developing a unique system for measuring Parkinson's disease tremors and gait analyses.
- ❑ The goal of this study is to provide cost-efficient and effective solutions based on basic approaches while maintaining personal comfort using mountable sensors such as gyro sensors and EMG sensors.
- ❑ The gait analysis sensor used shows to be successful in gathering the required data. Using the IMU accelerometer and gyroscope, various gait metrics such as stride time, stance time, step time, swing time, step length, gait speed, and stride length may be retrieved from the created system.
- ❑ When compared to people with abnormalities, the suggested approach produces consistent repeated flexion and extension of joints at standard gait, which is a major outcome.
- ❑ An adequate analytical approach may be developed to act as an acceptable rehabilitation strategy for patients with gait dysfunctions. A user-friendly interface can assist in sending data to the appropriate physicians to support effective management during monitoring.



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THANK YOU!